



01. MongoDB Architecture & Shell Fundamentals



What is MongoDB? (NoSQL Revolution)

MongoDB ek **open-source, document-oriented NoSQL database** hai. Isme data traditional tables aur rows ke badle **JSON-like flexible documents** mein store hota hai.



SQL (Relational) vs NoSQL (Document)

Feature	SQL (MySQL, PostgreSQL)	NoSQL (MongoDB)
Data Model	Strict Tables, Rows & Columns	Flexible JSON/BSON Documents
Schema	Fixed Schema (pre-defined columns)	Dynamic Schema (fields har document me alag ho sakte hain)
Relationships	JOINS & Foreign Keys	Embedded Documents ya References
Scaling	Vertical Scaling (bada server)	Horizontal Scaling (Sharding / clusters)



MongoDB Shell (mongosh) Essential Commands

```

mongosh - Terminal

# 1. Shell me enter karein
> mongosh

# 2. Saare existing databases dekhne ke Liye
> show dbs

# 3. Database me switch karne ya naya create karne ke Liye
> use college

# 4. Currently active database check karne ke Liye
> db
  
```



LAZY CREATION: use college likhne se database disk par tab tak create nahi hota jab tak pehla document insert na kiya jaye!



BSON vs JSON Deep Dive & First Document Insert



Master Comparison: BSON vs JSON (Top Interview Favorite!)

Feature	JSON (JavaScript Object Notation)	BSON (Binary JSON)
Format	Plain text (UTF-8 string)	Binary-encoded format
Readability	Human-readable (padh sakte hain)	Machine-readable (binary bytes)
Data Types	Limited: String, Number, Boolean, Array, Object, null	Rich: ObjectId, Date, BinData, Int32, Int64, Decimal128, Regex
Speed & Traversal	Slow parsing (string scan karna padta hai)	Fast traversal (har field ka length prefix store hota hai)
Storage Overhead	Text representation causes space overhead	Optimized for fast indexing & storage on disk



Anatomy of _id: ObjectId("65c8a1f2...")

MongoDB automatically har document ko 12-byte unique identifier assign karta hai:



Inserting First Document (insertOne)

```

mongosh - insertOne

// student collection me document insert karo
db.student.insertOne({
  name: "Rahul Sharma",
  age: 20,
  marks: 94,
  city: "Delhi"
});
// Return: { acknowledged: true, insertedId: ObjectId("65c8a1f2...") }
// Ab 'show collections' Likhne par 'student' display hoga!
  
```



EXAM ALERT: BSON me length prefixes ki wajah se database ko pura document parse nahi karna padta, wo directly agle field par jump kar sakta hai!